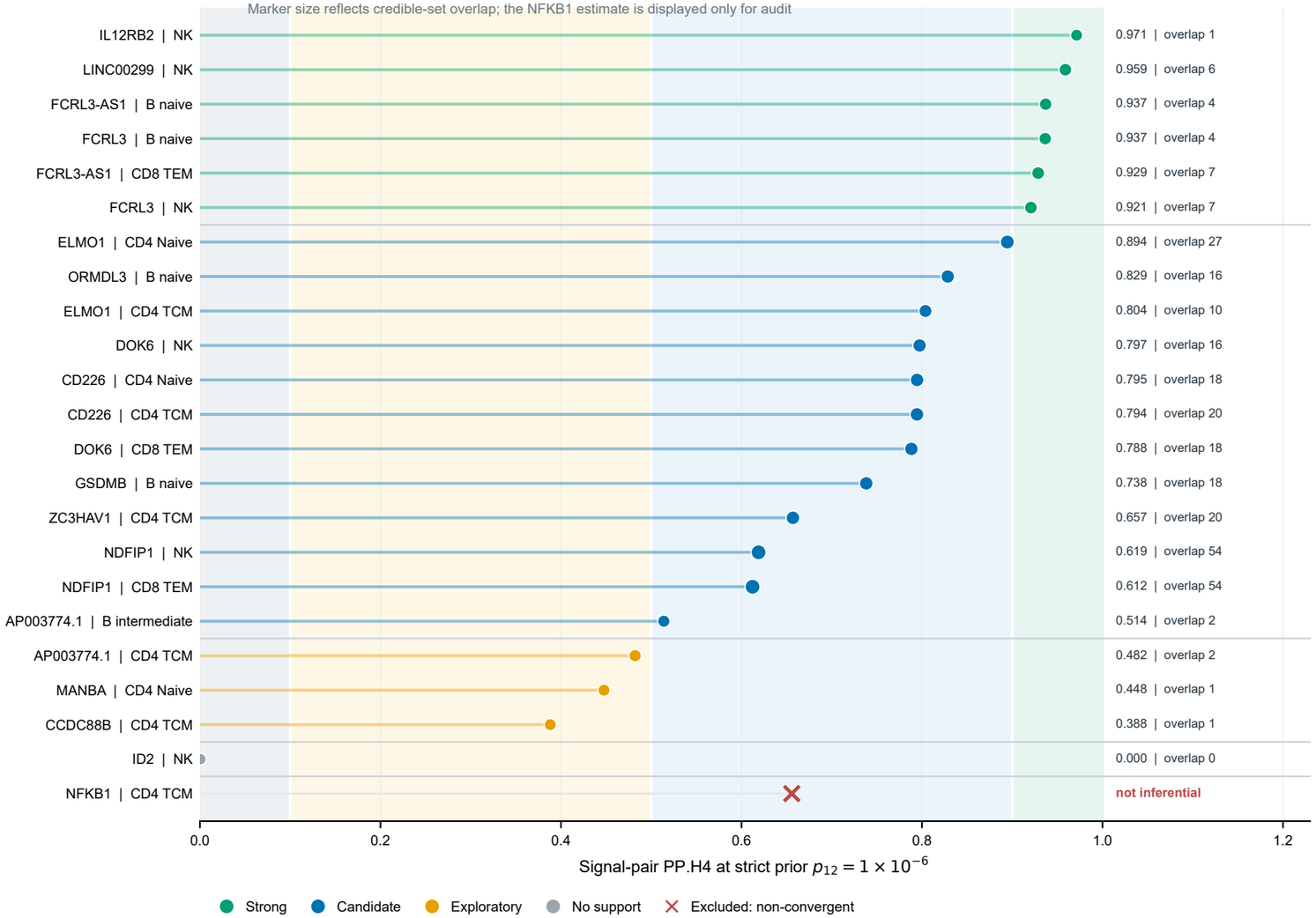
Screen-wide signal-level colocalization and model quality control

All 23 prespecified gene–cell combinations are shown; no result was removed because of statistical direction

A Complete prespecified discovery screen

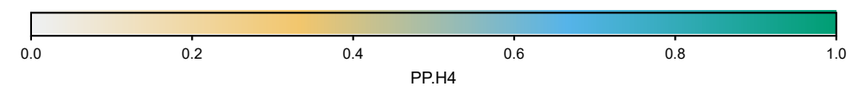
Marker size reflects credible-set overlap; the NFKB1 estimate is displayed only for audit



B Prior sensitivity across all combinations

Relaxed priors can upgrade candidates; final classes use the strict prior

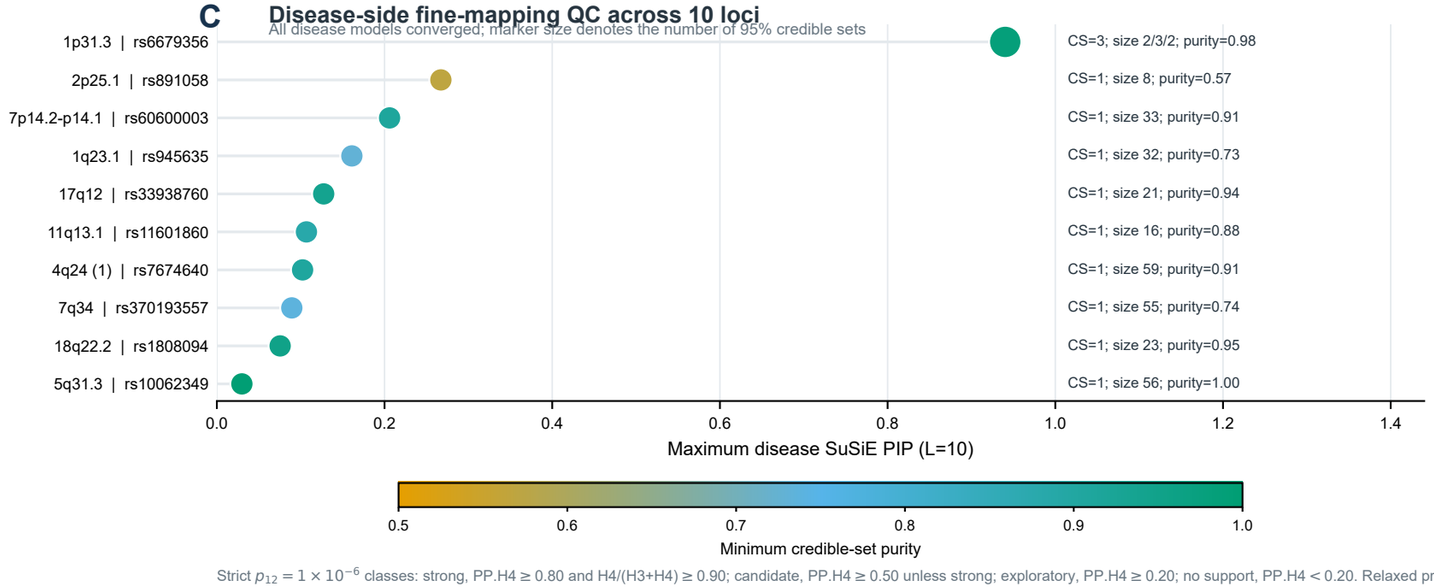
	1×10^{-6} strict	5×10^{-6}	1×10^{-5} relaxed
SA09 IL12RB2	0.97	0.99	1.00
SA03 LINC00299	0.96	0.99	1.00
SA07 FCRL3-AS1	0.94	0.99	0.99
SA05 FCRL3	0.94	0.99	0.99
SA08 FCRL3-AS1	0.93	0.98	0.99
SA06 FCRL3	0.92	0.98	0.99
SA01 ELMO1	0.89	0.98	0.99
SA10 ORMDL3	0.83	0.96	0.98
SA02 ELMO1	0.80	0.95	0.98
SA16 DOK6	0.80	0.95	0.98
SA14 CD226	0.79	0.95	0.97
SA13 CD226	0.79	0.95	0.97
SA15 DOK6	0.79	0.95	0.97
SA11 GSDMB	0.74	0.93	0.97
SA24 ZC3HAV1	0.66	0.91	0.95
SA17 NDFIP1	0.62	0.89	0.94
SA18 NDFIP1	0.61	0.89	0.94
SA22 AP003774.1	0.51	0.84	0.91
SA21 AP003774.1	0.48	0.82	0.90
SA20 MANBA	0.45	0.80	0.89
SA23 CCDC88B	0.39	0.76	0.86
SA04 ID2	0.00	0.00	0.00
SA19 NFKB1	x	x	x



NFKB1 values arise from a non-convergent eQTL model and are crossed out rather than interpreted.

C Disease-side fine-mapping QC across 10 loci

All disease models converged; marker size denotes the number of 95% credible sets



D Convergence audit and prespecified exception handling

A failed model is excluded unless a lower-complexity review converges

Disease SuSiE 20 / 20
L=3 and L=10 models

eQTL SuSiE L=10 21 / 23
two models reviewed at L=3

L=3 review 1 / 2
IL12RB2 rescued; NFKB1 failed

Final usable 22 / 23
one model excluded

SA09 IL12RB2 / NK L10 failed → L3 converged → INCLUDED

SA19 NFKB1 / CD4 TCM L10 failed → L3 failed → EXCLUDED

Interpretation lock: PP.H4 from a non-convergent model is an audit trace, not biological evidence.

Strict $p_{12} = 1 \times 10^{-6}$ classes: strong, PP.H4 ≥ 0.80 and H4/(H3+H4) ≥ 0.90; candidate, PP.H4 ≥ 0.50 unless strong; exploratory, PP.H4 ≥ 0.20; no support, PP.H4 < 0.20. Relaxed priors cannot upgrade a class; same-locus combinations are not independent discoveries.